Water Research 39 (2005) 4287–4298

WATER RESEARCH

www.elsevier.com/locate/watres

Water quality parameter estimation in a distribution system under dynamic state

G.R. Munavalli[1], M.S. Mohan Kumar*

Department of Civil Engineering, Indian Institute of Science, Bangalore 560 012, India

Received 21 August 2003; received in revised form 23 March 2005; accepted 27 July 2005

Abstract

Chlorine maintenance in distribution systems is an issue for water suppliers. The complex pipe geometry in distribution systems, the dynamic flow conditions experienced within them, and the varied nature of chlorine's reactivity make it difficult to predict chlorine levels throughout a water system. Computer-based mathematical models of water quality transport and fate within distribution systems offer a promising tool for predicting chlorine in a cost-effective manner. Nevertheless, the use of water quality models can only be effective and reliable when both hydraulics and the mechanisms of chlorine dissipation within the water system are properly defined. Bulk water decay can be measured experimentally. However, wall reaction rates are more complex to determine and must be deduced from field measurement by comparison with simulation results. The simulation–optimization model presented in this paper provides an effective tool to simplify the chlorine decay model calibration process that is often tedious. The optimization tool is based on the weighted-least-squares method solved by Gauss–Newton technique. Application of the model onto a real-life system shows that quantity, quality and location of measurement nodes play an important role in estimation of parameters.

Keywords: Chlorine decay modeling; Distribution systems; Drinking water; Optimization tool; Wall reaction parameter

1. Introduction

The study of water quality aspects within a municipal water distribution system is of great significance as it plays an important role in assuring a good quality of water to the consumer. The maintenance of residual chlorine is used as key criteria to assess the potability of water in the chlorine fed distribution systems. The spread of chlorine within the distribution system can be best studied by the use of mathematical models due to the complexities arising out of varying hydraulic conditions and non-applicability of universal chlorine reaction kinetics. The spatio-temporal variations in chlorine levels are established using forward simulation water quality model. The reaction parameters constitute a vital component of the input data needed for the realistic simulations using this model. The predicted chlorine concentrations within a distribution system are governed by reaction parameters (classified as overall, bulk and wall). However, the task of assigning these parameters to the pipes individually, globally or zoned is critical. It is generally assumed that chlorine decay in the bulk water entering the distribution system can be described by a

*Corresponding author.
E-mail addresses: gurumunavalli@yahoo.co.in (G.R. Munavalli), msmk@civil.iisc.ernet.in (M.S. Mohan Kumar).
[1]Presently, Faculty, Department of Civil Engineering, Walchand College of Engineering, Sangli 416 415, India.

0043-1354/$ - see front matter
doi:10.1016/j.watres.2005.07.043

first-order kinetic model. But the bulk decay parameter can also be non first-order and usually determined experimentally by the bottle test. The overall and wall reaction parameters are difficult to measure in the field. Hence the best way of estimating these parameters is through calibration against the field measurements.

Previously, the trial and error procedure (Clark et al., 1995) was used as a calibration approach. But this is too tedious and may not result in the proper estimation of the parameter values. Also very few studies have been reported in the literature on methodical estimation of specific parameters. Zeirolf et al. (1998) illustrated the use of input–output model for chlorine transport to estimate the first-order (global and zoned) wall reaction parameter. The model is applicable only for first-order reaction kinetics, and does not incorporate storage tanks and multiple water quality sources. Al-Omari and Chaudhry (2001) used finite difference procedures for the determination of overall first-order chlorine decay coefficient(s). Munavalli and Mohan Kumar (2003) developed an inverse model, which estimates the various reaction parameters in a multi-source steady-state distribution system. This model is extended herein for parameter estimation under dynamic state. It essentially has the autocalibration procedure consisting of simulation–optimization technique. The Lagrangian forward simulation water quality model is utilized in the model. The developed model, which computes the various reaction parameters in a more direct fashion, is memory efficient, free from numerical diffusion within the length of a segment and avoids the time-consuming trial and error procedure. In the following sections verification, applicability and usefulness of the inverse model are illustrated using real-life distribution systems.

2. Mathematical model

2.1. Forward simulation water quality model

The forward simulation model consists of hydraulic and chlorine transport components which are discussed below.

2.1.1. Hydraulic model

In the present study, the static hydraulic model (of Niranjan Reddy, 1994) is further modified to compute dynamic flows in the pipes using extended period simulation.

2.1.2. Chlorine transport model

When chlorinated water enters the distribution system, chlorine residual tends to dissipate. Three factors that frequently influence chlorine consumption are: (1) reactions with organic and inorganic chemicals (e.g., ammonia, sulfides, ferrous ion, manganous ion, humic material) in the bulk aqueous phase; (2) reactions with biofilm at the pipe wall; and (3) consumption by the corrosion process (Clark, 1998). Chlorine decay in distribution systems is generally considered to be composed of two components viz bulk and wall demands. The chlorine transport model is formulated assuming one-dimensional advection-dominated transport phenomenon within a pipe segment. Thus, the general governing equation for transport of chlorine along the ith pipe is given by

$$\frac{\partial c_{i,t}}{\partial t} = -u_i \frac{\partial c_{i,t}}{\partial x} - R(c_{i,t}), \tag{1}$$

where $c_{i,t}$ is the chlorine concentration in pipe i (mg/L) as a function of distance x and time t; u_i the mean flow velocity in pipe i (m/s) and $R(c_{i,t})$ the reaction rate expression.

$R(c_{i,t})$ represents the combined effect of bulk and wall reactions. The most common combination used is the first-order bulk and first-order wall reaction models. However in this paper, the purpose is to provide a large choice of kinetic models and are discussed in the following paragraphs.

(1) Overall first-order reaction kinetics

$$R(c_{i,t}) = -k_0 c_{i,t}, \tag{2}$$

where k_0 is the overall first-order reaction parameter (d^{-1}).

(2) First-order bulk and first-order wall reactions

$$R(c_{i,t}) = -k_{\mathrm{b},1i} c_{i,t} - \frac{k_{\mathrm{w},1i} k_{\mathrm{f}i}}{r_{\mathrm{h}i}(k_{\mathrm{w},1i} + k_{\mathrm{f}i})} c_{i,t}, \tag{3}$$

where, $r_{\mathrm{h}i}$ is the hydraulic radius (m); $k_{\mathrm{w},1i}$ the first-order wall reaction parameter (m/d); $k_{\mathrm{b},1i}$ the first-order bulk decay parameter (d^{-1}) and $k_{\mathrm{f}i}$ the mass transfer coefficient (m/d), the expression for the estimation of this parameter is described in detail by Rossman (2000).

(3) First-order bulk and zero-order wall reactions

$$R(c_{i,t}) = -k_{\mathrm{b},1i} c_{i,t} - \mathrm{Min}\left(\frac{k_{\mathrm{f}i} c_{i,t}}{r_{\mathrm{h}i}}, \frac{k_{\mathrm{w},0i}}{r_{\mathrm{h}i}}\right), \tag{4}$$

where $k_{\mathrm{w},0i}$ is the zero-order wall reaction parameter ($\mathrm{mg/m^2\,d}$).

(4) Second-order bulk reaction with respect to chlorine only and first-order wall reaction

$$R(c_{i,t}) = -k_{\mathrm{b},2i} c_{i,t}^2 - \frac{k_{\mathrm{w},1i} k_{\mathrm{f}i}}{r_{\mathrm{h}i}(k_{\mathrm{w},1i} + k_{\mathrm{f}i})} c_{i,t}, \tag{5}$$

$k_{\mathrm{b},2i}$ is the second-order bulk decay parameter (L/mg d).

(5) Second-order bulk reaction with respect to chlorine only and zero-order wall reaction

$$R(c_{i,t}) = -k_{\mathrm{b},2i} c_{i,t}(c_{i,t} - C_{\mathrm{L}}) - \frac{k_{\mathrm{w},0i} k_{\mathrm{f}i}}{r_{\mathrm{h}i}(k_{\mathrm{w},1i} + k_{\mathrm{f}i})} c_{i,t}. \tag{6}$$

(6) Two component second-order bulk decay and first-order wall reactions

$$R(c_{i,t}) = -k_{b,2i}c_{i,t}(c_{i,t} - C_L) - \frac{k_{w,1i}k_{fi}}{r_{hi}(k_{w,1i} + k_{fi})}c_{i,t}, \tag{7}$$

where C_L is the limiting concentration of chlorine (mg/L). The value C_L indicates that the chlorine will decay to a minimum limiting level and no further. The non-availability of sufficient organic matter to react with all the chlorine may be the reason for C_L.

(7) Two component second-order bulk and zero-order wall reactions

$$R(c_{i,t}) = -k_{b,2i}c_{i,t}(c_{i,t} - C_L) - \mathrm{Min}\left(\frac{k_{w,0i}}{r_{hi}}, \frac{k_{fi}}{r_{hi}}c_{i,t}\right). \tag{8}$$

The mixing of fluid is considered to be complete and instantaneous at the nodes where two or more pipes meet. Water leaving such nodes will carry the chlorine concentration, which is equal to the flow-weighted average of concentrations coming from the incoming pipes. Thus the instantaneous and complete mixing at the node j and time t_j is given by the equation

$$\mathrm{Cnc}_{j,t_j} = \frac{\sum_{i=1}^{N_{\mathrm{inp}j}} Q_i c_{i,t} + Q_E C_E}{\sum_{i=1}^{N_{\mathrm{inp}j}} Q_i + Q_E}, \quad j = 1, \ldots N_{jn} \tag{9}$$

where $N_{\mathrm{inp}j}$ is the number of incoming pipes at node j N_{jn} the total number of nodes in the network, Q_E the external source flow into at node j (m^3/s), and C_E the external source concentration into at node j (mg/L).

The above chlorine transport formulation is solved numerically by Lagrangian time-driven-method (Liou and Kroon, 1987) and thus the forward simulation model is developed. Input parameters needed for this model are either an overall first-order reaction parameter or a bulk reaction parameter along with the wall reaction parameters depending upon the type of reaction rate expressions used.

2.2. Simulation–optimization model

As previously mentioned the purpose is to estimate the global (same for all the pipes in the system) or zoned (multiple) wall reaction parameter(s) from the field chlorine measurements. For the zoned parameters, the grouping can be based on the age, roughness, material, diameter and flow of the pipe. The parameter estimation can be formulated as an optimization problem where the objective is to minimize some function of the difference between observed and model predicted responses.

2.2.1. Objective function

The objective function is given by

$$\text{Minimize} \quad E = \sum_{j=1}^{M}\sum_{k=1}^{N(j)} W_{j,k}[\mathrm{Cno}_{j,t_k} - \mathrm{Cnc}_{j,t_k}]^2, \tag{10}$$

where $\mathrm{Cno}_{j,tk}$ is the observed value of chlorine concentration at node j and time t_k (mg/L), $\mathrm{Cnc}_{j,tk}$ the computed value of chlorine concentration at node j and time t_k (mg/L), $W_{j,k}$ the weight associated with node j and time t_k, and $N(j)$ the number of observations available at node j and M the number of observation nodes.

2.2.2. Solution technique

The optimization problem is solved by using Gauss–Newton sensitivity analysis technique the details of which are given in Munavalli and Mohan Kumar (2003).

2.2.3. Calibration error statistics

The statistical parameters need to be computed to assess accuracy of the fitting between observed and simulated concentrations. The parameters computed are total number of observations, mean of observed values, mean of computed values, mean error, root mean square (RMS) error, and correlation between means, for each measurement location and for network as a whole. As the simulation time in the chlorine transport model is advanced by a water quality time step, it happens that the observation time may not coincide with the simulation time. Hence it is essential to interpolate the computed chlorine concentration corresponding to the observation time. In the present study, the required concentration is linearly interpolated using the concentrations of end simulation times.

2.2.4. Estimation of parameter uncertainty

Information concerning the uncertainty in parameter estimates is contained in the posteriori parameter covariance matrix (Mishra and Parker, 1989). The covariance matrix of estimated parameters also provides information regarding the reliability of each of these parameters. A well-estimated parameter is generally characterized by a small variance as compared to an insensitive parameter that is associated with a large variance. The more sensitive the parameter, the closer and quicker the parameter will converge. A correlation analysis of the estimated parameters would indicate the degree of interdependence among the parameters with respect to the objective function. A first-order approximation of the covariance matrix is given by Bard (1974)

$$C \approx s^2(\mathbf{J}_f^T \mathbf{W} \mathbf{J}_f)^{-1}. \tag{11}$$

Here $\mathbf{J}_f$ is the final Jacobian of sensitivity matrix and s^2 is the estimated error variance

$$s^2 = \frac{E}{\left(\sum_{j=1}^{M} N(j) - N_{up}\right)} \tag{12}$$

where N_{up} is the number of unknown parameters. The diagonal elements of the covariance matrix C contain the individual parameter variances, while the off diagonal elements reflect the correlation between parameters. Individual parameter confidence limits can also be determined from the parameter variances and the appropriate value of Student's t distribution with $(M-N_{up})$ degrees of freedom. Parameter confidence limits are given by

$$P_r[k_{wt} - tC_{ll}^{1/2} \leq k_{wl} \leq k_{wl} + tC_{ll}^{1/2}] = 1 - \alpha, \tag{13}$$

where k_{wl} is the estimated value of lth parameter, C_{ll} the parameter variance, t the value of the Student's t distribution for confidence level $(1-\alpha)$, and $(M-N_{up})$ degrees of freedom, which can be obtained from the inverse t distribution (Abramowitz and Stegun, 1964). Although the statistical measures of parameter reliability discussed above are strictly applicable for linear regression, they have also been shown to be reasonably accurate in nonlinear least-squares problems (Donaldson and Schnabel, 1986). In the present study, the 95% confidence interval value (i.e., $\alpha = 0.05$) is used to quantify the uncertainty in the parameter estimates.

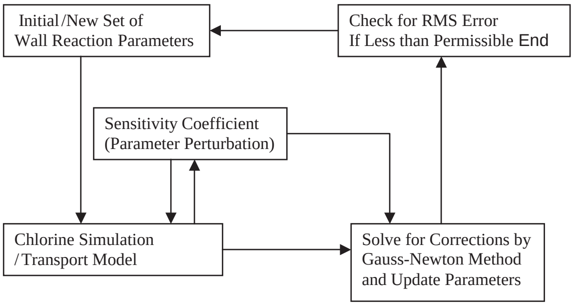

Fig. 1. Simulation–optimization methodology.

2.2.5. Choice of weights

The weight based on the mean of the observed concentrations, which was found to perform better

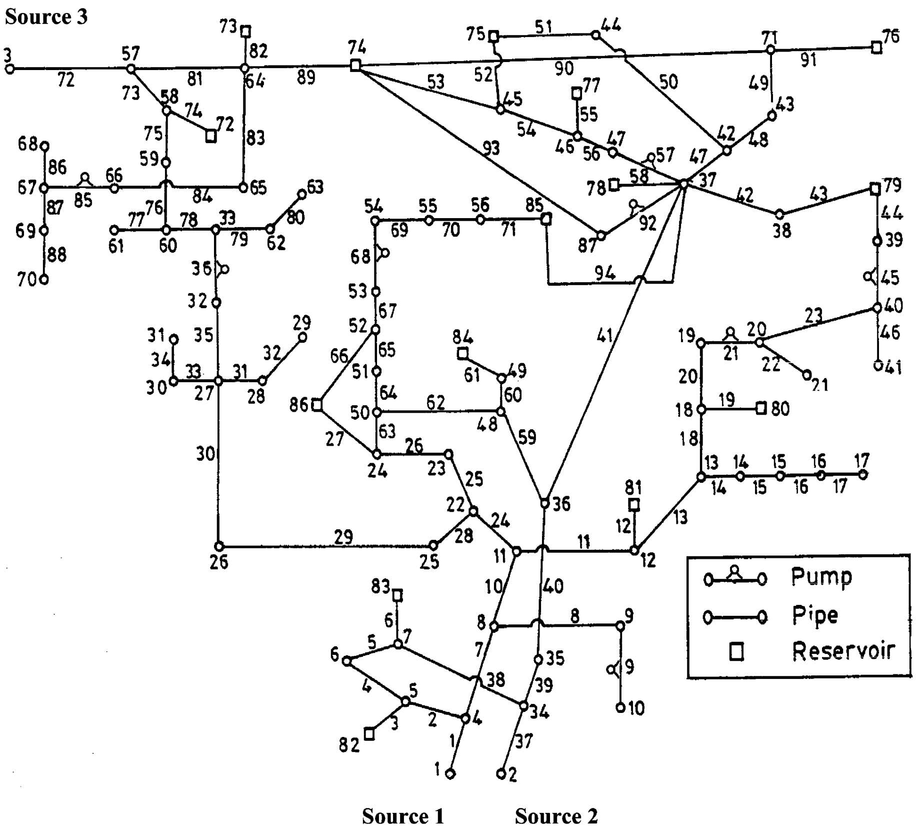

Fig. 2. Verification network example.

Table 1
Demand patterns for verification network example

Period (h)	Hourly demand flow patterns (m^3/s)								
	4	5	6	7	8	9	10	11	12
1	0	0	0	0	0	0	0	0	0
2	0	0	0	0	0	0	0	0	0
3	0	0	0	0	0	0	0	0	0
4	0	0	0	0	0	0	0	0	0
5	0	0	0	0	0	0	0	0	0
6	0	0	0	0	0	0	0	0	0
7	0.053	0.042	0.189	0.179	0.2	0.21	0.021	0.074	0.158
8	0.053	0.042	0.189	0.179	0.2	0.21	0.021	0.074	0.158
9	0.053	0.042	0.189	0.179	0.2	0.21	0.021	0.074	0.158
10	0.053	0.042	0.189	0.179	0.2	0.21	0.021	0.074	0.158
11	0.053	0.042	0.189	0.179	0.2	0.21	0.021	0.074	0.158
12	0.053	0.042	0.189	0.179	0.2	0.21	0.021	0.074	0.158
13	0	0	0	0	0	0	0	0	0
14	0	0	0	0	0	0	0	0	0
15	0	0	0	0	0	0	0	0	0
16	0	0	0	0	0	0	0	0	0
17	0	0	0	0	0	0	0	0	0
18	0	0	0	0	0	0	0	0	0
19	0.053	0.042	0.189	0.179	0.2	0.21	0.021	0.074	0.158
20	0.053	0.042	0.189	0.179	0.2	0.21	0.021	0.074	0.158
21	0.053	0.042	0.189	0.179	0.2	0.21	0.021	0.074	0.158
22	0.053	0.042	0.189	0.179	0.2	0.21	0.021	0.074	0.158
23	0.053	0.042	0.189	0.179	0.2	0.21	0.021	0.074	0.158
24	0.053	0.042	0.189	0.179	0.2	0.21	0.021	0.074	0.158

Table 2
Nodes with demand patterns (verification network example)

Pattern	Node
1	10, 17, 21, 29, 41, 61, 63
2	68, 70
3	31
4	30, 6
5	14, 16
6	28
7	33
8	44
9	54
10	55, 69
11	60
12	62

Table 3
Input chlorine concentrations (verification network example)

Node	Time (h)	Chlorine concentration (mg/L)
73	4	0.61
74	12	0.56
75	5	0.32
76	20	0.26
78	18	0.36
80	4	0.35
82	10	0.62
84	21	0.53
86	14	0.49

(Munavalli and Mohan Kumar, 2003), is applied in the present study.

Weights based on mean of measured values is given by

$$W_{j,k} = \left[\frac{1}{\mathrm{Cno}_{\mathrm{mean}}}\right]^2, \tag{14}$$

where $\mathrm{Cno}_{\mathrm{mean}}$ is the mean of the measurements and $W_{j,k}$ is constant for all measurements.

2.2.6. *Solution methodology*

The connection between simulation and optimization tool is represented in the Fig. 1.

3. Model verification

The procedure adopted for verification of simulation–optimization model consists of two steps. In the first step, synthetic measurement data (location, time and concentration) are generated by running the forward simulation water quality model with the assumed reaction parameters. In the second step, initial estimates of the reaction parameters and synthetic measurements constitute input data for the model. The model run should yield the true reaction parameters if it is said to be verified.

3.1. Verification network example

The water mains network of Bangalore city in India is used for model verification and is shown in Fig. 2. The details of the network are given in Datta and Sridharan (1994). The demand patterns 1, 2, and 3 have constant hourly demands of 0.0631, 0.021 and 0.0315 m^3/s throughout the simulation period of 24 h. Table 1 shows the other demand patterns applied onto the network. The demand patterns associated with the various nodes are given in Table 2.

A nonlinear chlorine reaction kinetics consisting of second-order bulk and first-order wall reaction is assumed to be applicable for the network. The second-order bulk reaction parameter used is 4.5 L/mg d. The entire network is divided into three groups of pipes according to their periodwise association with three sources of supply. The pipes 72–94, 1–36, and 37–71 form three groups and are assigned with wall reaction parameters 0.75, 0.50 and 1.0 m/d, respectively. The three sources of supply have a constant input chlorine concentration of 0.75 mg/L. Following the verification procedure mentioned above, the chlorine concentrations are established at all nodes by running the forward simulation water quality model. Then, chlorine concentrations at the nodes given in Table 3 are assumed to be input measurements for the simulation–optimization model and the reaction parameters, that are considered to be unknown, are estimated. The model is run by using various initial (over-, under- and mixed-) estimates of parameters. Figs. 3(a)–(c) show the rapid convergence of the parameter values to their respective true values for all the three initial estimates used. Fig. 3(d) shows the convergence of objective function value to zero.

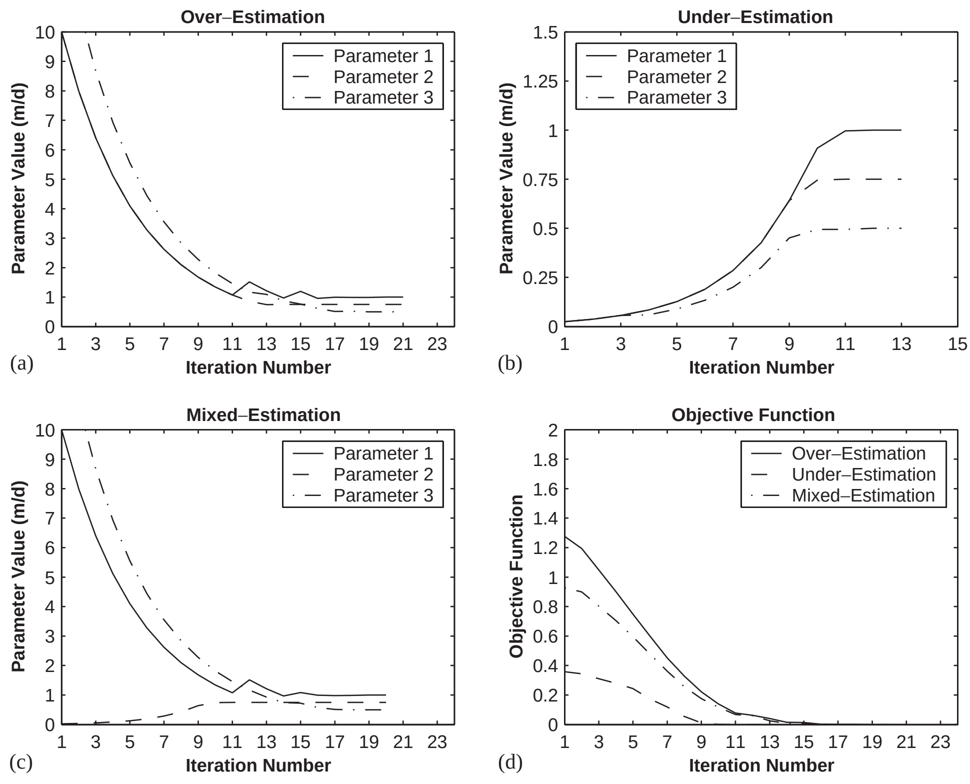

Fig. 3. Convergence results: (a) parameter 1, (b) parameter 2, (c) parameter 3, and (d) objective function value for verification network example.

4. Model application: results and discussion

In this section, the model is applied to estimate the unknown wall reaction parameters of two water distribution systems for which the measured chlorine data are available in the literature. The two systems used in the present study are Brushy plains zone (Connecticut) and Fairfield zone 3 (California).

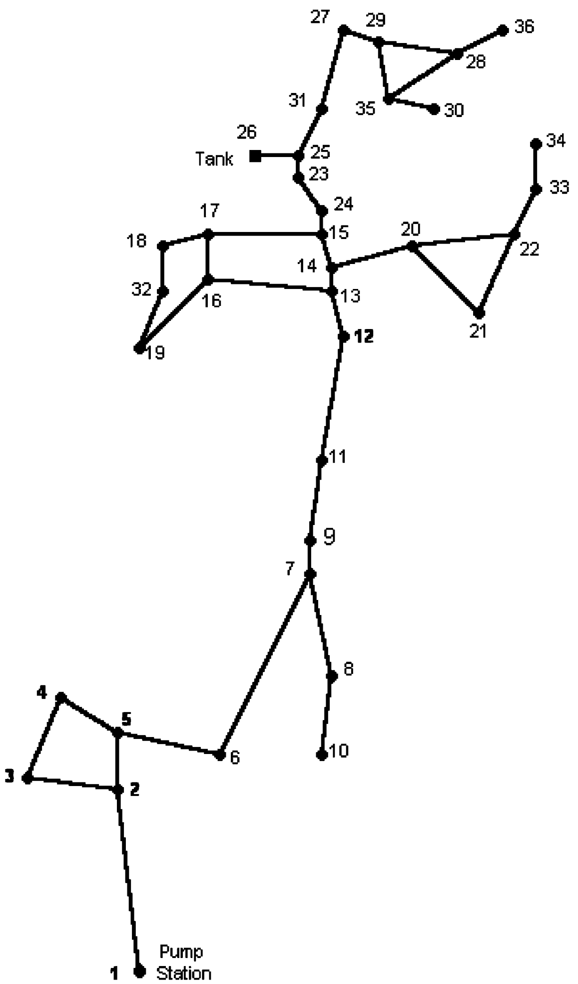

Fig. 4. Application network example 1.

4.1. Application network example 1

The Brushy plains water distribution system, which is shown in Fig. 4, is used as an application network 1. The details of the network are adopted from example problem of EPANET distributed by USEPA (Rossman, 2000). The chlorine input to the network has a constant value of 1.15 mg/L and the bulk first-order parameter used throughout the network is 0.55 d^{-1}. The in situ chlorine concentrations at the nodes 3, 6, 10, 11, 19, 25, 28 and 34 observed for a period of 55 h adopted from Rossman et al. (1994) are used as input to the model. The system is analyzed for two cases of parameter estimation viz. global and zoned. In the former case, a single (global) reaction parameter is assigned for the entire network whereas in the latter case the wall parameter is assigned for the group of pipes based on the diameters. A zero concentration tolerance, 3 min quality step, 1 h hydraulic step and 55 h total simulation period are used for both these cases.

4.1.1. Global parameter estimation

In this case the model is run to estimate the global parameters k_0, $k_{w,1}$ and $k_{w,0}$. The results, which include parameter value, confidence band and objective function value, are presented in Table 4. The estimated value of k_0 equal to 2.5169 d^{-1} which is considerably larger than the bulk parameter signifying major contribution from wall to the total chlorine decay. The other parameters ($k_{w,1}$ and $k_{w,0}$) estimated give an idea of how much is the wall contribution and which is the dominating reaction kinetics for the system. The calibration statistics for accuracy of the fitting is shown in Table 5 for all the parameters computed above. It can be seen from Table 5 that the statistical variation is marginal between the first-order and zero-order wall reaction models. The calibration of water quality model for this system carried out using trial and error procedure by Rossman et al. (1994) reported a range of global first-order wall reaction parameter ($k_{w,1}$) values between 0.15 and 0.45 m/d. The lower value (0.15 m/d) of the above range corresponds to a RMS residual error of 0.192 mg/L whereas the upper value (0.45 m/d) has RMS error of 0.175 mg/L. The model presented in this study estimated directly the value of this parameter to be 0.3654 m/d,

Table 4
Parameters estimated for application network example 1

Parameter	Estimated value	Unit	Confidence band		Objective function value
			Minimum	Maximum	
k_0	2.5169	d^{-1}	2.1482	2.8856	32.35
$k_{w,1}$	0.3654	m/d	0.2790	0.4517	25.14
$k_{w,0}$	201.61	$mg/m^2/d$	159.48	243.74	25.65

Table 5
Calibration statistics for accuracy of the fitting (application network example 1)

Parameter	Unit	Number of observations	Observed mean (mg/L)	Computed mean (mg/L)	Mean error (mg/L)	RMS error (mg/L)	Correlation between means
k_0	d^{-1}	155	0.45	0.44	0.14	0.19	0.96
$k_{w,1}$	m/d	155	0.45	0.43	0.13	0.17	0.99
$k_{w,0}$	$mg/m^2/d$	155	0.45	0.43	0.13	0.17	0.99

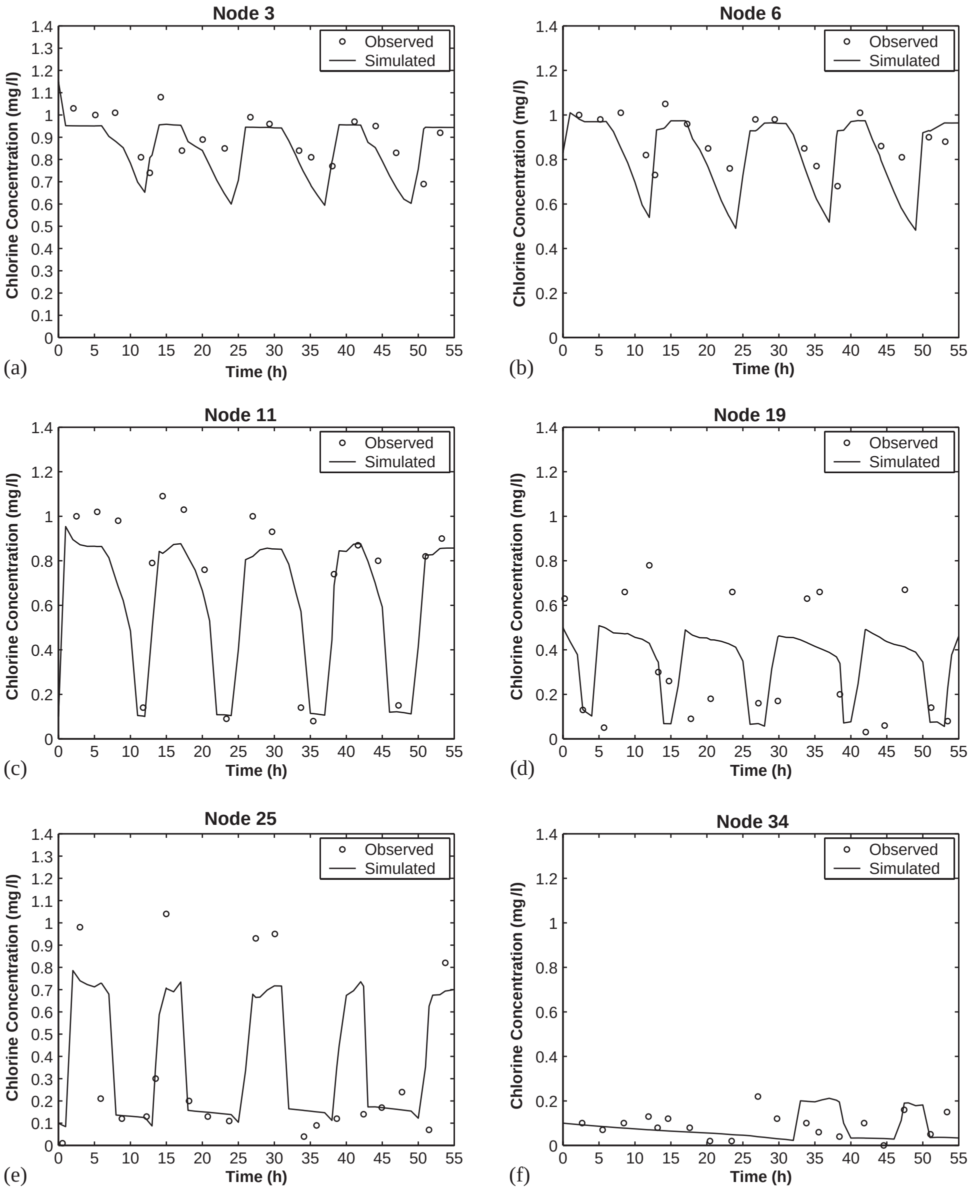

Fig. 5. Computed and observed chlorine concentrations at: (a) node 3, (b) node 6, (c) node 11, (d) node 19, (e) node 25, and (f) node 34 for application network example 1.

which corresponds to the lowest possible RMS residual error of 0.172 mg/L without any tedious trial and error computations. The comparison of observed and simulated chlorine concentrations, which are obtained using first-order bulk and first-order wall reaction kinetics at node 3, 6, 11, 19, 25 and 34 are represented in Fig. 5. It

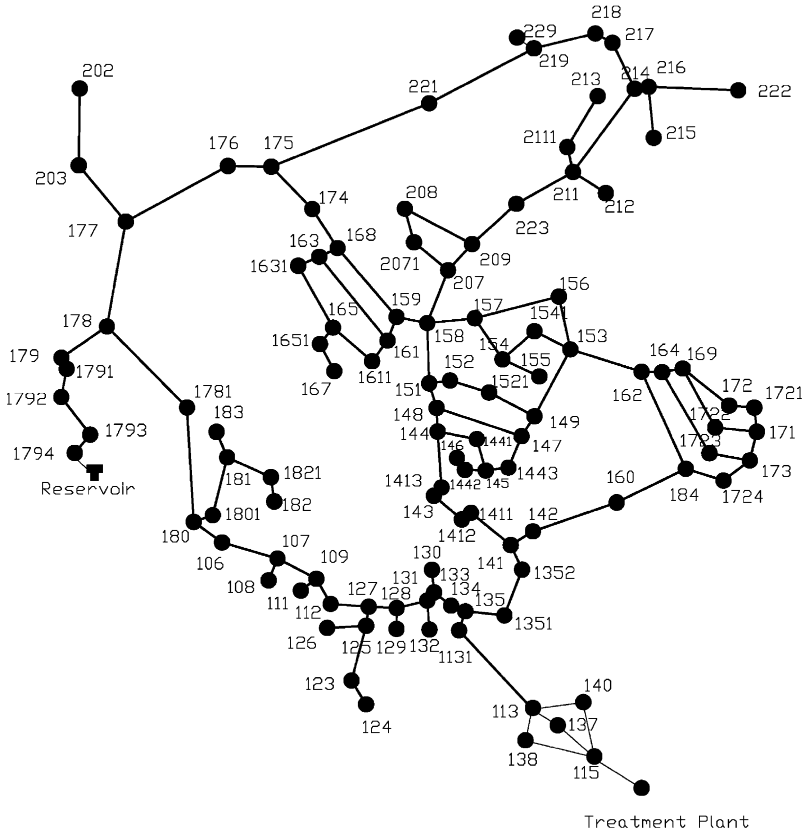

Fig. 6. Application network example 2.

Table 6
Calibration statistical results for application network example 2

Location	Number of observations	Observed mean (mg/L)	Computed mean (mg/L)	Mean error (mg/L)	RMS error (mg/L)
106	10	1.21	1.15	0.15	0.32
134	10	1.43	1.47	0.05	0.06
141	9	1.37	1.44	0.09	0.12
144	7	1.37	1.40	0.05	0.05
1521	9	0.99	0.92	0.19	0.26
153	9	1.04	0.99	0.10	0.14
158	9	1.25	1.17	0.10	0.12
162	9	1.06	1.04	0.11	0.16
168	9	0.89	0.97	0.16	0.31
172	9	1.02	1.07	0.16	0.19
178	15	0.75	0.94	0.34	0.47
1821	8	0.88	0.81	0.10	0.12
184	9	1.13	1.20	0.08	0.09
214	9	0.70	0.66	0.10	0.13
219	9	0.31	0.27	0.13	0.17
Network	140	1.01	1.03	0.14	0.20

Correlation between means: 0.971. Overall reaction parameter: $1.25\,d^{-1}$. Objective function value: 7.27. Parameter 95% confidence limits: $1.04–1.47\,d^{-1}$.

can be seen from figure that the simulated chlorine concentrations do not match well with the observed ones at some nodes. To match the peaks at these nodes the lower wall reaction parameter is to be used and vice versa. But the global parameter estimated by the present model provides the best possible fit for the observations at all the monitoring nodes. The discrepancy between the simulated and observed values for this system are due to the frequent flow reversals in the pipes, very long residence time in some pipes due to low water demands, large hydraulic time step and high residence time in the storage tank. This discrepancy has also been reported in the past by Rossman et al. (1994) and Ozdemir and Ucak (2002) while using different water quality modeling technique.

4.1.2. Zoned parameter estimation

To study the variation of wall reaction parameters with respect to the diameter, the system is zoned into two groups having pipe diameter of 0.3048 and 0.2032 m, respectively. The model is run using the first-order bulk and first-order wall reaction kinetics. And the

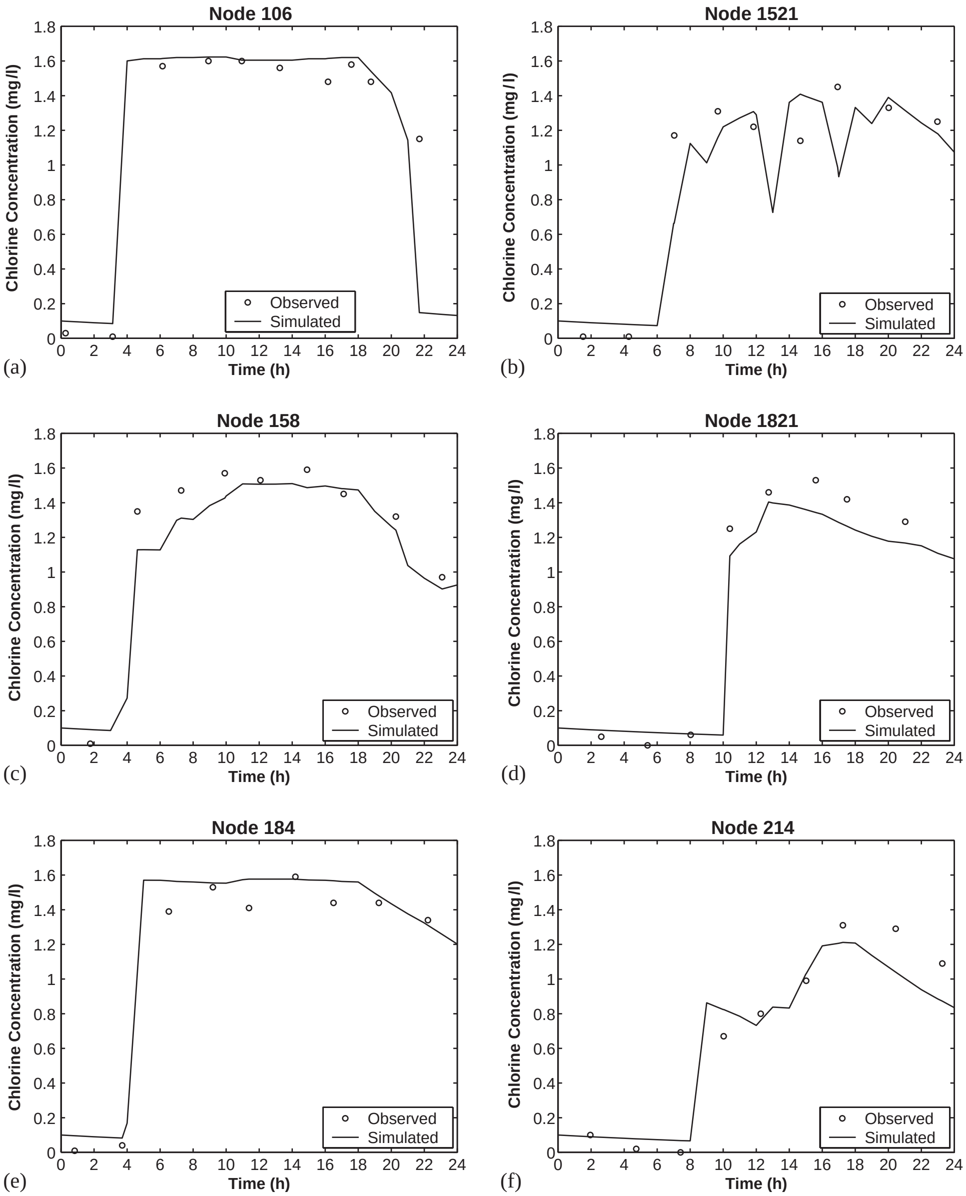

Fig. 7. Computed and observed chlorine concentrations at nodes: (a) 106, (b) 1521, (c) 158, (d) 1821, (e) 184, and (f) 214 for application network example 2.

values of estimated wall reaction parameters are 0.3308 and 0.5014 m/d for the two groups, respectively. The results show that the group with larger diameter has a smaller value of wall reaction parameter and vice versa. It indicates that the contribution from the wall reactions is more in smaller diameter pipes to the overall decay than the larger diameter pipes. This observation is similar to the one reported by Rossman et al. (1994).

4.2. Application network example 2

The second network used in this study is the water system for city of Fairfield-zone 3 and is shown in Fig. 6. The network has 126 pipes, 111 nodes, one storage reservoir and one treatment plant. The system consists of relatively new large–diameter asbestos–cement pipes, hence the network is assigned with a first-order bulk reaction parameter of $1.16\,d^{-1}$ (Vasconcelos et al. (1994)). The chlorine concentration of 1.70 mg/L was introduced into the system at the treatment plant and periodic chlorine-residual measurements were taken at various locations. The details of location and time of chlorine-residual measurements are obtained from L. A. Rossman, through personal communication.

The model is run to determine the global overall first-order reaction parameter. The estimated value of the parameter is $1.25\,d^{-1}$, which is slightly higher than the bulk parameter indicating the lesser contributions from wall reactions. To confirm and quantify this observation the model is run for the cases of first-order and zero-order wall reactions, and the corresponding reaction parameters are determined to be negligibly small. The calibration error statistics are presented in Table 6 for the parameter estimated above. The observed and simulated chlorine concentrations for nodes 106, 1521, 158, 1821, 184 and 214 are represented in Fig. 7. There is a good agreement between the observed and simulated chlorine concentrations. These results are similar to those observed by Vasconcelos et al. (1997).

5. Conclusions

The simulation–optimization model is formulated and solved for parameter estimation using the weighted-least-squares method based on Gauss–Newton minimization technique. The model utilizes the forward simulation model in its computational domain. It can estimate the parameters involved in overall first-order, first- and zero-order wall reaction kinetics. The parameter perturbation technique is used for evaluating the sensitivity coefficients.

The model verification illustrates the capability of the model to handle non-linear chlorine reaction kinetics with rapid convergence to the correct values. The application to real-life water distribution systems shows usefulness of the model to compute the various kinds of global and zoned reaction parameters. The key of model is the avoidance of previously used tedious trial–error approach for parameter estimation. Further, the computation of calibration statistics and parameter uncertainty help to select the suitable chlorine reaction kinetics and adopt appropriate parameters for the system. The model presented provides a good tool for the water supply authorities to calibrate the water quality model, following either a first-order or non-first-order chlorine reaction kinetics, for their system.

Acknowledgement

The writers would like to acknowledge Dr. Lewis. A. Rossman of the US EPA for providing the necessary data on the networks discussed in the paper.

References

Abramowitz, M., Stegun, I.A., 1964. Handbook of Mathematical Functions. Dover, Mineola, NY 1046pp.

Al-Omari, A.S., Chaudhry, M.H., 2001. Unsteady-state inverse chlorine modeling in pipe networks. Journal of Hydraulic Engineering, ASCE 127 (8), 669–677.

Bard, Y., 1974. Nonlinear Parameter Estimation. Academic Press, San Diego, CA 341pp.

Clark, R.M., 1998. Chlorine demand and TTHM formation kinetics: a second order model. Journal of Environmental Engineering, ASCE 124 (1), 16–24.

Clark, R.M., Rossman, L.A., Wymer, L.J., 1995. Modeling distribution system water quality: regulatory implications. Journal of Water Resources Planning and Management, ASCE 121 (6), 423–428.

Datta, R.S.N., Sridharan, K., 1994. Parameter estimation in water-distribution systems by least squares. Journal of Water Resources Planning and Management, ASCE 120 (4), 405–422.

Donaldson, J.R., Schnabel, R.S., 1986. Computational experience with confidence regions and confidence intervals for nonlinear least squares. In: Allen, D.M. (Ed.), Computer Science and Statistics. Elsvier Science, New York, pp. 83–86.

Liou, C.P., Kroon, J.R., 1987. Modeling the propagation of waterborne substances in water distribution networks. Journal AWWA 79 (11), 54–58.

Mishra, S., Parker, J.C., 1989. Parameter estimation for coupled unsaturated flow and transport. Water Resources Research 25 (3), 385–396.

Munavalli, G.R., Mohan Kumar, M.S., 2003. Water quality parameter estimation in a steady state distribution system. Journal of Water Resources Planning and Management, ASCE 129 (2), 124–134.

Niranjan Reddy, P.V., 1994. General analysis, parameter estimation and valve operational policy in water distribution networks. Ph.D. Thesis. Indian Institute of Science, Bangalore.

Ozdemir, O.N., Ucak, A., 2002. Simulation of chlorine decay in drinking-water distribution systems. Journal of Environmental Engineering, ASCE 128 (1), 31–39.

Rossman, L.A., 2000. EPANET 2 Users Manual. Risk Reduction Engineering Laboratory, US Environmental Protection Agency, Cincinnati, OH.

Rossman, L.A., Clark, R.M., Grayman, W.M., 1994. Modeling chlorine residuals in drinking water distribution systems. Journal of Environmental Engineering, ASCE 120 (4), 803–820.

Vasconcelos, J.J., Grayman, W.M., Boulos, P.F., Clark, R.M., Rossman, L.A., Goodrich, J.A., 1994. Characterization and modeling of chlorine decay in water distribution networks. In: Proceedings of AWWA 1994 Annual Conference, Denver, CO.

Vasconcelos, J.J., Rossman, L.A., Grayman, W.M., Boulos, P.F., Clark, R.M., 1997. Kinetics of chlorine decay. Journal AWWA 89 (7), 55–65.

Zeirolf, M.L., Polycarpou, M.M., Uber, J.G., 1998. Development and autocalibration of an input–output model of chlorine transport in drinking water distribution systems. IEEE Transactions on Control Systems Technology 6 (4), 543–553.